Networking

ASSIGNMENT

BASIC COMMUNICATION HARDWARE

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Introduction to basic communication hardware.

Communication hardware is a set of physical or network devices that are essential for interaction and communication between hardware units operational on a computer network.

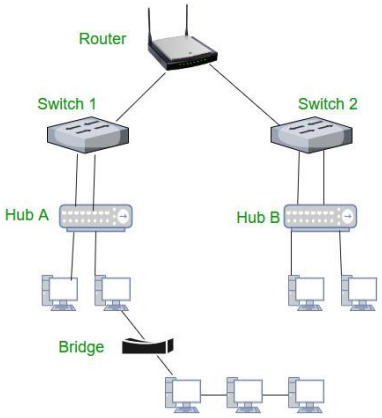
Basic data communication hardware includes the physical devices that are used to transmit and receive data between different devices these devices include :

1: network interface card (NIC). 2: switch

2: cable 4: router

5: hub 6: bridge

7: modem 8: firewall



So We are going to discuss one by one by those communication devices that we expressed above

1: Network Interface Card (NIC).

also known as a Network Adapter or LAN Adapter, is a hardware component that provides a physical connection between a computer or other device and a local network. The NIC allows the device to communicate with other devices, including computers, servers, printers, and other peripherals that are connected to the same network.



The function of a NIC is to provide a physical interface between a device and a network. It receives digital signals from a device, converts them into a format compatible with the network, and transmits them over the network. Similarly, it receives signals from the network, converts them into a format that the device can understand, and sends them to the device.

The NIC works by receiving electrical signals from the device's motherboard and converting them into signals that can be transmitted over a network cable. When it receives data from the network, it converts it into a format that the device can understand.

There are several types of NICs available, including:

1. Ethernet NIC: The most common type of NIC used today, Ethernet NICs are used to connect devices to a local area network (LAN). They operate at speeds ranging from 10 Mbps to 10 Gbps.

2. Wi-Fi NIC: These are used to connect devices to a wireless network. They support various wireless standards such as 802.11a, b, g, n, and ac.

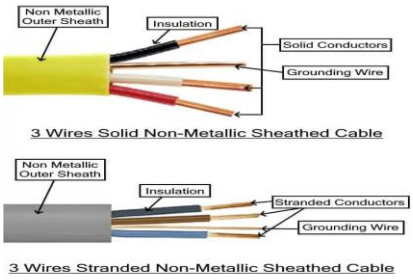
3. Fiber Optic NIC: These are used to connect devices to a fiber optic network. They operate at very high speeds and are commonly used in data centers.

4. Token Ring NIC: These are used in older networks and are similar to Ethernet NICs but use different network protocols.

NB:The type of NIC needed depends on the type of network being used and the requirements of the device being connected.

2: Cables.

are physical wires that connect one electronic device to another or connect a device to a network. They serve as a conduit through which electric signals can pass. The function of a cable is to transmit signals or electrical power from one device to another.



Cables work by carrying electrical energy from one point to another. They have conductive materials inside, such as copper wires, which allow electric current to flow easily. The current travels along the wires, and the signals or data are transmitted to or from the electronic devices. There are many types of cables, including:

1. Coaxial Cable: This cable has a central conductor surrounded by a layer of insulation, a braided shield, and an outer insulating layer. It is commonly used to transmit cable TV signals and broadband internet.

2. Twisted Pair Cable: This cable consists of two or more insulated wires twisted around each other. It is commonly used for Ethernet networks and telephone systems.

3. Fiber Optic Cable: This cable consists of thin glass fibers that carry signals using light waves. It is commonly used for high-speed data transmission and long-distance communication

4. USB Cable: USB (Universal Serial Bus) cables are used to connect devices such as printers, cameras, and external hard drives to computers. They come in various types, including USB-A, USB-B, and USB-C.

5. HDMI Cable: HDMI (High-Definition Multimedia Interface) cables are used to connect devices such as TVs, Blu-ray players, and gaming consoles to display devices. They transmit high-definition audio and video signals. The type of cable needed depends on the specific requirements of the devices being connected and the type of signal being transmitted.

3: switch

is a network device that connects devices on a computer network. It is used to connect devices such as computers, servers, printers, and other hardware on a wired or wireless network. The primary function of a switch is to receive, process, and forward data packets between network devices.



The working of a switch involves the following steps:

1. When a device wants to send data to another device on the network, it sends a data packet to the switch.

2. The switch receives the packet and checks the destination MAC address of the packet to determine which port the packet should be sent out to reach the device.

3. The switch sends the packet out only to the port on which the destination device is connected.

4. The switch also maintains a table of all connected devices and their corresponding MAC addresses for efficient packet forwarding.

Types of Switches:

1. Unmanaged Switches: These are basic switches that allow devices to connect to each other without any network configuration required. They are ideal for small networks, and they are typically less expensive than managed switches.

2. Managed Switches: These switches offer more advanced features such as port monitoring, traffic prioritization, VLANs (Virtual Local Area Networks), Quality of Service (QoS), and security protocols. They are ideal for large networks with complex requirements.

3. Smart Switches: These switches offer some of the advanced features of a managed switch but without the complexity. They are ideal for small to medium-sized networks with basic network requirements.

4. PoE Switches: These switches provide Power over Ethernet (PoE) capabilities, which allow devices such as IP cameras, wireless access points, and VoIP phones to receive power and data over a single Ethernet cable.

5. Layer 3 Switches: These switches offer routing capabilities in addition to switching, allowing them to route traffic between different subnets on a network.

Choosing the right type of switch depends on the specific network requirements and the number and type of devices that need to be connected.

4: router.

is a network device that connects multiple networks and forwards data packets between them. It is mainly used to connect a local area network (LAN) to the internet or a wide area network (WAN).

Function of a Router: The primary function of a router is to forward data packets between networks.



Working of a Router:

When a device sends a data packet to another device, it first sends it to the default gateway, which is the router. The router examines the destination IP address of the packet and checks its routing table to determine the best path for the packet to reach its destination.

Types of Routers:

1. Wired Router: A wired router is a network device that connects wired devices to a network, such as computers and servers, using ethernet cables.

2. Wireless Router: A wireless router is a network device that connects wireless devices to a network using Wi-Fi technology. It can be used to connect smartphones, laptops, and other mobile devices to the internet or a local network.

3. Core Routers: These are high-performance routers used in large networks such as internet service providers or backbone networks.

4. Edge Routers: These are smaller routers used to connect to branch offices or remote locations.

5. Virtual Routers: These are software routers that run on a virtual machine and can be used to create virtual private networks (VPNs) or test network configurations.

Choosing the right type of router depends on the size and complexity of the network, the number of devices that need to be connected, and the required features such as security and performance.

5: hub.

is a networking device that connects multiple devices together in a network. It is a simple device that operates at the Physical Layer (Layer 1) of the OSI model.



Function of a Hub: The primary function of a hub is to connect multiple devices together to create a network.

Working of a Hub:

When a device sends a data packet to other devices in a network, the hub receives the packet and broadcasts it to all devices connected to it.

Types of Hubs:

1. Passive Hub: A passive hub does not have the capability to amplify or regenerate signals, it simply connects multiple devices together to create a network.

2. Active Hub: An active hub is an advanced version of a passive hub that has the capability to amplify the signal and regenerate it before sending it to other devices. It also has additional features such as collision detection and filtering.

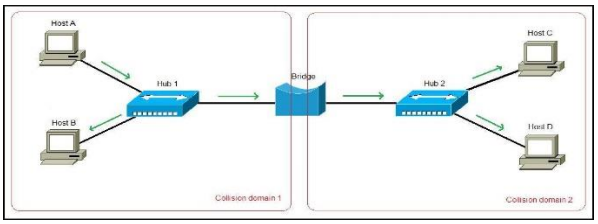
3. Switching Hub: A switching hub is an advanced hub that is capable of filtering data packets based on their destination addresses. It forwards the packet only to the designated device or port where the device is connected, making the network more efficient.

6: bridge.

is a networking device that connects multiple network segments together and can selectively forward data between them. It is a device that operates at the Data Link Layer (Layer 2) of the OSI model.

Function of a Bridge:

The primary function of a bridge is to connect two or more network segments together and selectively forward data between them. Bridges operate at the Data Link Layer and use hardware addresses (MAC addresses) to determine where to forward the data packets.





Working of a Bridge:

When a data packet enters a bridge, the bridge examines the destination MAC address of the packet to determine which segment the packet should be forwarded to. If the destination MAC address is on the same segment as the source, the bridge will not forward the packet to any other segment. If the destination address is on a different segment, the bridge will forward the packet to that segment.

Types of Bridges:

1. Local Bridge: A local bridge connects two or more network segments within the same physical location.

2. Remote Bridge: A remote bridge connects two or more network segments in different physical locations using a Wide Area Network (WAN).

3. Wireless Bridge: A wireless bridge is a bridge that connects two or more wireless networks, allowing them to communicate with each other.

4. Transparent Bridge: A transparent bridge is a bridge that forwards packets without modifying them, making it appear as if the packets are transmitting directly between the connected segments.

5. Source Routing Bridge: A source routing bridge is a bridge that forwards packets according to the pre-determined path specified in the packet header.

Choosing the right type of bridge depends on the network topology and the required functionality. However, with the advancement of technology and the rise of more efficient devices such as switches and routers, bridges are no longer commonly used in modern networks.

7: modem.

is a device that translates digital signals from a computer or other digital devices into analog signals that can be transmitted over a phone line and vice versa. Modem is short for Modulator-Demodulator and is used to connect devices to the internet through a telephone line.

Function of a Modem:

The primary function of a modem is to modulate digital data from a computer into analog signals so that it can be transmitted over a phone line and then received by a distant modem, which demodulates the signal to convert it back to digital data that can be read by the receiving computer. This allows computers to communicate with each other via the phone company's infrastructure.



Working of a Modem:

When a user wants to connect to the internet, the digital signal from the computer is converted into analog signals by the modem and transmitted over a phone line. The receiving modem demodulates the signal to convert it back to a digital signal, which can then be received by the recipient's computer.

Types of Modems:

1. Dial-up Modems: A dial-up modem is the simplest and slowest type of modem which uses analog signals over a regular telephone line to connect to the internet. Dial-up modems work at a speed of 56 kbps.

2. DSL Modems: A DSL modem, also known as Digital Subscriber Line modem, uses digital signals over a regular telephone line to connect to the internet. DSL modem provides faster internet speed than a dial-up modem and works at a speed between 128 kbps to 100 Mbps.

3. Cable Modems: A cable modem connects to the internet through the local cable TV provider and provides faster internet speed than both dial-up and DSL modems. Cable modems provide speed between 1 Mbps to 1 Gbps.

4. Wireless Modems: A wireless modem is similar to a normal modem, but it connects to the internet using cellular networks. Wireless modems work at a speed between 2 Mbps to 20 Mbps.

5. Satellite Modems: A satellite modem uses satellite technology to connect to the internet. This is the slowest and most expensive type of modem and is used in remote areas where other types of internet connections are not possible.

Choosing the right modem depends on the type of internet connection available in the area and the internet speed required by the user.

8: firewall.

is a software or hardware device that monitors and controls incoming and outgoing network traffic based on predetermined security rules. It acts as a barrier between a computer network and the outside world, providing a layer of protection against unauthorized access and threats.



Function of a Firewall:

The primary function of a firewall is to prevent unauthorized access to a computer network by monitoring and controlling incoming and outgoing network traffic. It provides a secure boundary between the network and the internet, preventing malicious traffic from entering the network and mitigating the spread of malware, viruses, and other types of cyber-attacks.

Working of a Firewall:

Firewalls work by analyzing incoming and outgoing traffic based on defined rules and policies. It checks the network traffic and identifies the source and destination of the traffic, the type of traffic, and the routing information. Based on the defined rules, the firewall either blocks or allows the traffic.

Types of Firewall:

1. Packet Filtering Firewall: A packet filtering firewall examines incoming and outgoing network packets based on a set of predetermined filtering rules. It checks the source and destination address, protocol, and port number of each packet and allows or denies it based on the filtering rules.

2. Stateful Inspection Firewall: A stateful inspection firewall keeps track of the state of each network connection and examines the traffic based on the context of the communication. It

determines whether incoming packets are part of an existing connection by examining the packet headers and matching them to existing connections.

3. Proxy Firewall: A proxy firewall acts as an intermediary between the internal and external networks. It examines network traffic at the application layer and can inspect the contents of the communication, including the payload and metadata.

4. Next-generation Firewall: Next-generation firewalls use advanced techniques such as deep packet inspection, intrusion prevention, and threat intelligence to provide more comprehensive security and protection against advanced cyber attack

9: conclusion

In conclusion, basic data communication hardware is essential for transmitting data between devices. It includes several components such as

cables, connectors, routers, switches, hubs, and modems.

All of the above components work together to ensure that data can be transmitted effectively and efficiently.

The choice of hardware depends on factors such as the distance over which data needs to be transmitted, the speed at which it needs to be transmitted, the number of devices that need to be connected, and the security requirements of the network. Understanding basic data communication hardware is crucial for setting up a reliable and secure network .

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